

INSTALLATION CASE STUDY

Vaillant aroTHERM plus 12 kW | R290 Air Source Heat Pump Retrofit

Clean Energy. Smarter Future.



The Right Spot Is Everything.

A period cottage, a tight courtyard and an R290 heat pump that has to sit somewhere compliant, quiet and out of the way. Siting designed first, system designed around it.

SYSTEM

aroTHERM plus 12 kW

REFRIGERANT

R290 propane

PROPERTY

Period cottage

INSTALLER

Elixia Renewables

THE BRIEF

A period cottage, a tight courtyard, and nowhere obvious to put it

The hardest part of most heat pump installations is not the heat pump. It is finding a position for it that is compliant, quiet, efficient and something the owner is happy to look at every day.

This property is a rendered stone cottage taking a 12 kW unit, with a paved courtyard on the sheltered side and neighbouring properties close by. The usable outdoor space is generous to look at and unforgiving to design around: walls, steps, planting, drainage and windows all competing for the same few square metres.

Three constraints pulling against each other

Refrigerant safety. The unit specified runs on R290 - propane. That brings excellent efficiency and high flow temperatures, but it also brings a defined protective zone around the outdoor unit. Get the position wrong and the installation is not compliant, however tidy it looks.

A courtyard the owners actually use. This is a seating area, not a service yard. Whatever went in had to sit against the elevation and leave the paving, steps and planting exactly as they were.

Existing radiators. A period property with an established heating system. Ripping out every emitter to chase a low flow temperature was neither wanted nor necessary.

Siting is not the last decision in a heat pump design. On a constrained property it is the first one, and everything else is built around it.

The design envelope

Confirmed specification where stated. Heat loss, floor area and design flow temperature are indicative ranges typical of this class of system rather than measured values for this property.

Heat pump	Vaillant aroTHERM plus, 12 kW nominal output
Property	Detached period cottage, approx. 150-220 m2
Design heat loss	9-12 kW at -2 to -4 degrees C external
Emitters	Existing radiators, 45-55 degrees C design flow
Hot water	300 litre unvented cylinder with immersion back-up
Refrigerant	R290 propane, GWP of 3 against 675 for R32
Siting	Protective zone survey and full siting design

THE SOLUTION

Understanding the protective zone

R290 is propane, and propane is denser than air. That single fact governs how the protective zone behaves - and it is the reason this position works.

1 The zone projects outwards and downwards, not upwards

In the unlikely event of a leak, refrigerant sinks and disperses at low level. The manufacturer's protective zone therefore extends around and below the unit, not above it. A position that looks wrong at a glance can be entirely correct once the geometry is understood.

2 No openings or ignition sources within the zone

Windows, doors, light shafts, cellar entrances, flat-roof windows and ventilation openings must all fall outside it, as must ignition sources such as sockets, lamps and rotary isolators. Every one of those was surveyed and plotted before the base was laid.

3 Quiet enough to sit next to

The aroTHERM plus is Quiet Mark accredited, and in a sheltered courtyard it is unobtrusive in normal use. The position still earns that: solid wall behind, clear discharge across open paving, nothing for the fan to push against.

4 High flow temperature, existing radiators retained

R290 delivers usable flow temperatures well beyond what older refrigerants manage comfortably. That kept the existing emitters in service and the disruption outdoors, where it belongs.



Final position. Hard against the elevation, level on bearers, clear discharge across the courtyard, planting and paving undisturbed - and every opening on that wall outside the protective zone.

THE INSTALLATION

Tidy outside, undisturbed inside

The outdoor unit sits on solid bearers on the existing paving, with condensate managed away from the walking route and pipework kept short and insulated into the building. The courtyard planting, steps and seating area are exactly as they were.

Indoors, the cylinder and controls were installed with the existing radiator circuit retained and rebalanced to suit the new design flow temperature, with weather compensation set at commissioning rather than left on a factory default.

Scope of works

- Vaillant aroTHERM plus 12 kW air source heat pump, R290 refrigerant
- Protective zone survey and siting design prior to installation
- Full MCS siting and compliance assessment
- 300 litre unvented hot water cylinder with immersion back-up
- Existing radiator circuit retained, rebalanced and re-commissioned
- Weather compensation control strategy set at commissioning
- System flushing, inhibitor dosing, commissioning and handover

Typical performance for a system of this type

3.0-3.5

Seasonal efficiency for a well-designed system on existing radiators

Up to 75 C

Flow temperature available from R290, so existing emitters can stay

GWP 3

Global warming potential of propane, against 675 for R32

Typical ranges for this class of system, confirmed per project at design stage. Maximum flow temperature is the manufacturer's stated capability, not a recommended design condition - lower flow temperatures give better efficiency.

Tight sites are not a reason to say no to a heat pump. They are a reason to design the siting first and the rest of the system around it.

Been told your property is too tight for a heat pump?

Courtyards, side returns, terraces, listed and period properties - the positions that look impossible usually are not, once the protective zone is properly understood. We survey siting before we quote.

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